

Module classification in conformal field theory through symmetric polynomials

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Motivation

Bottom-up approach to constructing conformal field theories:

- Pick a chiral algebra (vertex operator algebra)
- Classify modules in a suitable category
- Compute characters and their modular properties
- Find modular invariants
- Compute fusion products of modules
- Compute correlation functions

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Today's chiral CFTs

The Virasoro chiral CFT $\text{Vir}_c, c \in \mathbb{C}$:

- 1 generating field $T = \sum_{n \in \mathbb{Z}} L_n z^{-n-2}$
- OPE relation: $T(z)T(w) = \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w} + \text{reg}$
- Commutation relations: $[L_m, L_n] = (m-n)L_{m+n} + \delta_{m+n,0} \frac{c}{12}(m^3 - m)$

The free boson $\mathcal{H}$:

- 1 generating field $a = \sum_{n \in \mathbb{Z}} a_n z^{-n-1}$
- OPE relation: $a(z)a(w) = \frac{1}{(z-w)^2} + \text{reg}$
- Commutation relations: $[a_m, a_n] = m\delta_{m+n,0}$
- Choice of conformal structure: $T(z) = \frac{1}{2} : a(z)^2 : + \frac{\alpha_0}{2} \partial a(z),$
 $c = 1 - 3\alpha_0^2$

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Vacuum Verma module structure of Vir_c

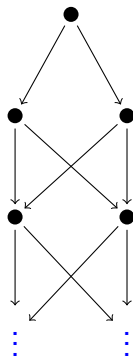
generic c



$c_{1,p}$



$c_{p,q}$



$$c_{p,q} = 1 - 6 \frac{(p-q)^2}{pq}, \quad \gcd(p, q) = 1.$$

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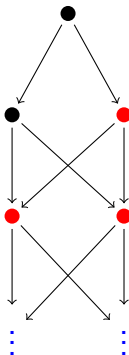
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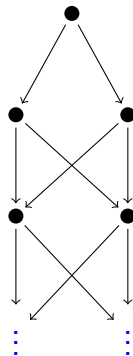


Assert state-field correspondence $\rightarrow L_{-1}\Omega = 0$.

Kac table, $c = -\frac{22}{5}, p = 2, q = 5$, Yang-Lee model

0	$-\frac{1}{5}$	$-\frac{1}{5}$	0	$\frac{2}{5}$	1	$\frac{9}{5}$
$\frac{11}{8}$	$\frac{27}{40}$	$\frac{7}{40}$	$-\frac{1}{8}$	$-\frac{9}{40}$	$-\frac{1}{8}$	$\frac{7}{40}$
4	$\frac{14}{5}$	$\frac{9}{5}$	1	$\frac{2}{5}$	0	$-\frac{1}{5}$
$\frac{63}{8}$	$\frac{247}{40}$	$\frac{187}{40}$	$\frac{27}{8}$	$\frac{91}{40}$	$\frac{11}{8}$	$\frac{27}{40}$
13	$\frac{54}{5}$	$\frac{44}{5}$	7	$\frac{27}{5}$	4	$\frac{14}{5}$

white



coloured



Module classification: Yang-Lee model

- Ideal of null vectors/fields generated by a vacuum module singular vector χ

$$\chi = (L_{-2}^2 - \frac{3}{5}L_{-4})\Omega$$
$$Y(\chi; z) = :T(z)^2: - \frac{3}{10}\partial^2 T(z) = \sum_{n \in \mathbb{Z}} \chi_n z^{-n-4}$$

- Evaluate χ_0 on a candidate Virasoro highest weight vector Ω_h .

$$0 = \chi_0 \Omega_h = (L_0^2 + \frac{1}{5}L_0)\Omega_h = h(h + \frac{1}{5})\Omega_h$$

- Conformal weight can only be 0 or $-\frac{1}{5}$.

Classifying minimal model modules

- Previous example was not a fluke. Up to some “technicalities” (codified in Zhu’s algebra), the module theory is completely characterised by the action of zero modes on highest weight vectors.
- Previous example depended heavily on explicit form of the singular vector. Singular vectors are **very** hard to find in general. No general closed formulae are known.
- Goal for the rest of the talk: Find a better way to evaluate zero modes of singular vectors on candidate highest weight vectors.

Zhu’s algebra:

- Zhu’s algebra is the associative algebra of zero modes acting on vectors in modules which are annihilated by positive modes.
- Constructible as a vector space quotient of V .
$$\text{Zhu}(V/I) = \text{Zhu}(V)/\text{Zhu}(I).$$
- Simple V -modules $\xleftrightarrow{1:1}$ simple Zhu-modules.

Yang-Lee using Zhu's algebra

- $\text{Zhu}(\text{Vir}_c) = \mathbb{C}[x]$ for all $c \in \mathbb{C}$, where $x = [T]$ is the image of the conformal vector.
- For $c = -\frac{22}{5}$, the image of the singular vector is $[\chi] = f(x) = x(x + \frac{1}{5})$.
- Explicit presentation of the minimal model Zhu algebra.
 $\text{Zhu}(\text{M}(2, 5)) = \frac{\mathbb{C}[x]}{\langle f(x) \rangle}$
- On simple modules over $\text{Zhu}(\text{M}(2, 5))$, x must act by multiplication by a zero of $f(x)$.

Module classification and symmetric polynomials

vertex operator algebras

ideals and quotients

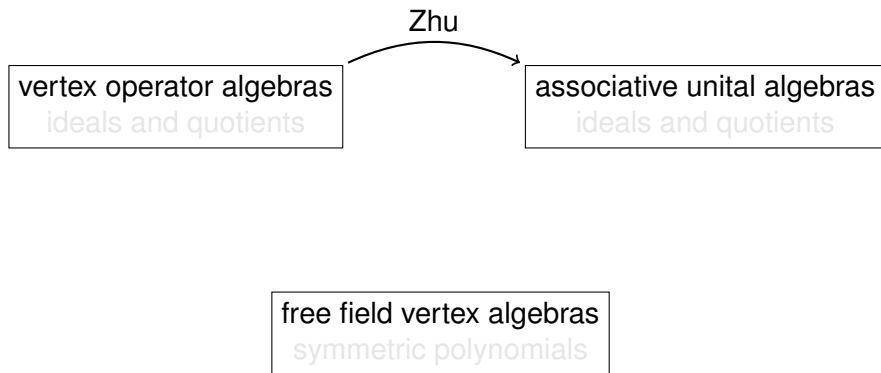
associative unital algebras

ideals and quotients

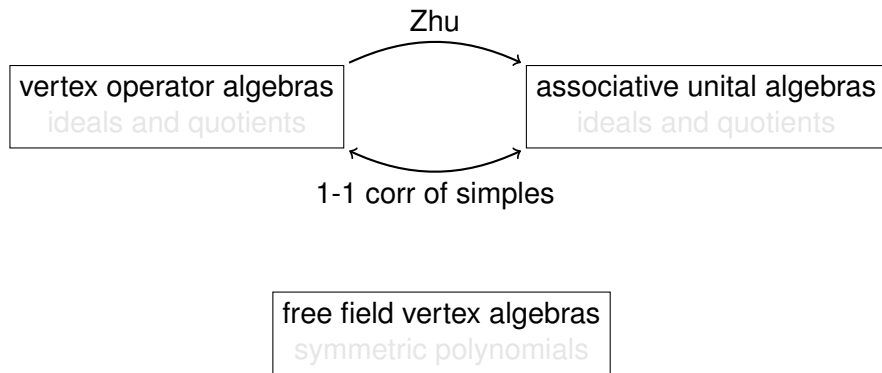
free field vertex algebras

symmetric polynomials

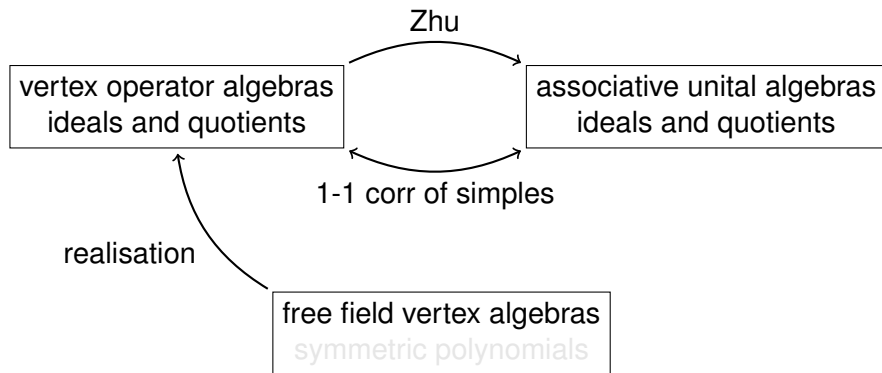
Module classification and symmetric polynomials



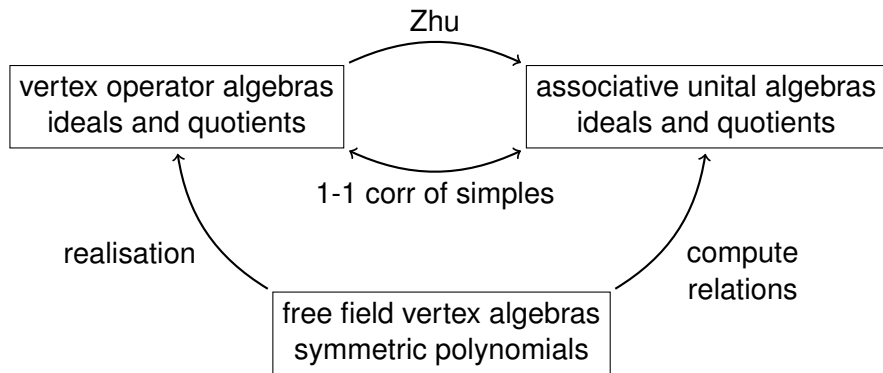
Module classification and symmetric polynomials



Module classification and symmetric polynomials



Module classification and symmetric polynomials



Free field realisation

Recall free boson CFT $\mathcal{H}$.

- Generated by $a(z)$.
- $T(z) = \frac{1}{2} : a(z)^2 : + \frac{\alpha_0}{2} \partial a(z)$, $c = 1 - 3\alpha_0^2$.
- Let $\phi = \hat{a} + \log(z)a_0 - \sum_{n \neq 0} \frac{a_n}{n} z^{-n}$, $\partial\phi = a$, $[a_m, \hat{a}] = \delta_{m,0}$.
- For α_+, α_- , such that $\alpha_+ \alpha_- = -2$ and $\alpha_+ + \alpha_- = \alpha_0$,

$$S_{\pm}(w) = : e^{\alpha_{\pm} \phi(w)} :$$

is a screening field: $Q_{\pm} = \text{Res}_{w=0} [S_{\pm}(w)]$ commutes with $T(z)$.

- $Q_{\pm}$ defines a map between Fock spaces $F_{\lambda} \rightarrow F_{\lambda+\alpha_{\pm}}$ whenever the residue is well defined.
- Since $Q_{\pm}$ commutes with $T(z)$, the image of the highest weight vector of F_{λ} will be **singular** in $F_{\lambda+\alpha_{\pm}}$.

The singular vector at minimal model central charges

The vacuum singular vector can only be constructed by composing multiple screening operators.

Let $p, q \geq 2$, $\gcd(p, q) = 1$ and

$$\alpha_+ = \sqrt{\frac{2q}{p}}, \alpha_- = -\sqrt{\frac{2p}{q}}, \alpha_0 = \alpha_+ + \alpha_-, \quad c = 1 - 3\alpha_0^2 = 1 - 6\frac{(p-q)^2}{pq}.$$

The vacuum singular vector in $\text{Vir}_c \subset \mathcal{H}$ is

$$\chi = Q_+^{[p-1]}|(1-p)\alpha_+\rangle \propto Q_-^{[q-1]}|(1-q)\alpha_-\rangle \in \text{Vir}_c \subset \mathcal{H}.$$

The minimal model conformal weights are the zeros of the action of the zero mode of χ on highest weight vectors.

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Composing screening operators

To construct the singular vector of $\text{Vir}_c \subset \mathcal{H}$, we must compose screening fields.

$$S_+(z_1) \cdots S_+(z_r) = e^{r\alpha_+ \hat{a}} \prod_{i \neq j} (z_i - z_j)^{\alpha_+^2/2} \prod_{i=1}^r z_i^{\alpha_+ + a_0} \\ \prod_{m \geq 1} \exp \left(\alpha_+ \frac{a_{-m}}{m} \sum_i z_i^m \right) \exp \left(-\alpha_+ \frac{a_m}{m} \sum_i z_i^{-m} \right)$$

Prop

If and only if $\mu = \frac{1-r}{2}\alpha_+ - \frac{k}{2}\alpha_-$, $k \in \mathbb{Z}$, there exists a unique choice of a cycle Γ such that

$$Q_+^{[r]} = \int_{\Gamma} S_+(z_1) \cdots S_+(z_r) dz_1 \cdots dz_r : F_{\frac{1-r}{2}\alpha_+ - \frac{k}{2}\alpha_-} \rightarrow F_{\frac{1+r}{2}\alpha_+ - \frac{k}{2}\alpha_-}$$

is a non-trivial Virasoro homomorphism.

- if $k \geq 0$ then $Q_+^{[r]} | \frac{1-r}{2}\alpha_+ - \frac{k}{2}\alpha_- \rangle = 0$.
- if $k < 0$ then $Q_+^{[r]} | \frac{1-r}{2}\alpha_+ - \frac{k}{2}\alpha_- \rangle \neq 0$.

Appearance of symmetric polynomials

Action of $Q_+^{[r]}$ on $F_{\frac{1-r}{2}\alpha_+ - \frac{k}{2}\alpha_-}$:

$$Q_+^{[r]} = e^{r\alpha_+ \hat{a}} \int_{\Gamma} \prod_{i \neq j} \left(1 - \frac{z_i}{z_j}\right)^{\alpha_+^2/2} \prod_{i=1}^r z_i^{k+1} \\ \times \prod_{m \geq 1} \exp\left(\alpha_+ \frac{a_{-m}}{m} \sum_i z_i^m\right) \exp\left(-\alpha_+ \frac{a_m}{m} \sum_i z_i^{-m}\right) \frac{dz_1 \cdots dz_r}{z_1 \cdots z_r}$$

This integral defines a well known inner product on the ring of symmetric polynomials $\mathbb{C}[z_1, \dots, z_r]^{\mathfrak{S}_r}$:

$$\langle f, g \rangle_r^t = \int_{\Gamma} \prod_{i \neq j} \left(1 - \frac{z_i}{z_j}\right)^t f(z_1^{-1}, \dots, z_r^{-1}) g(z_1, \dots, z_r) \frac{dz_1 \cdots dz_r}{z_1 \cdots z_r}$$

This inner product is symmetric and it vanishes if $\deg(f) \neq \deg(g)$.

Symmetric polynomials

Ring of symmetric polynomials in r variables: $\Lambda_r = \mathbb{C}[z_1, \dots, z_r]^{\mathfrak{S}_r}$

Many well known generators:

- Power sums

$$p_k(z) = \sum_i z_i^k$$

- Elementary symmetric polynomials

$$e_k(z) = \sum_{1 \leq i_1 < \dots < i_k} z_{i_1} \cdots z_{i_k}$$

$$\Lambda_r = \mathbb{C}[p_1, \dots, p_r] = \mathbb{C}[e_1, \dots, e_r]$$

Symmetric polynomials

Many well known bases. Generally indexed by partitions

$$\lambda = [\lambda_1, \lambda_2, \dots].$$

- Power sum basis: $\mathbf{p}_\lambda = \mathbf{p}_{\lambda_1} \mathbf{p}_{\lambda_2} \cdots$
- Elementary symmetric basis: $\mathbf{e}_\lambda = \mathbf{e}_{\lambda_1} \mathbf{e}_{\lambda_2} \cdots$
- Monomial symmetric polynomials:

$$\mathbf{m}_\lambda(z) = \sum_{\sigma} z_1^{\lambda_{\sigma(1)}} z_2^{\lambda_{\sigma(2)}} \cdots,$$

where the σ are all distinct permutations of $\lambda = [\lambda_1, \lambda_2, \dots]$.

$$\mathbf{m}_{[2,2]}(z_1, z_2) = z_1^2 z_2^2, \quad \mathbf{m}_{[2,2]}(z_1, z_2, z_3) = z_1^2 z_2^2 + z_1^2 z_3^2 + z_2^2 z_3^2.$$

$$\Lambda_r = \bigoplus_{\lambda, \lambda_1 \leq r} \mathbb{C} \mathbf{p}_\lambda = \bigoplus_{\lambda, \lambda_1 \leq r} \mathbb{C} \mathbf{e}_\lambda = \bigoplus_{\lambda, \ell(\lambda) \leq r} \mathbb{C} \mathbf{m}_\lambda$$

Jack symmetric polynomials

Jack symmetric polynomials $\mathbf{P}_\lambda^t(z)$ are uniquely characterised by

- 1 Upper triangular: $\mathbf{P}_\lambda^t(z) = \mathbf{m}_\lambda(z) + \sum_{\nu < \lambda} c_{\nu,\lambda}(t) \mathbf{m}_\nu(z)$
- 2 Orthogonal: $\langle \mathbf{P}_\lambda^t, \mathbf{P}_\nu^t \rangle_r^t = 0$, if $\lambda \neq \nu$.

Well studied:

- $\langle \mathbf{P}_\lambda^t(z), \mathbf{P}_\lambda^t(z) \rangle_r^t = a_\lambda(t)$
- $\prod_{i,j} (1 - y_i z_j)^{-t} = \prod_{m \geq 1} \exp\left(t \frac{\mathbf{p}_m(y) \mathbf{p}_m(z)}{m}\right) = \sum_\nu b_\nu(t) \mathbf{P}_\nu^t(y) \mathbf{P}_\nu^t(z)$
- $\prod_i (1 + z_i)^{Xt} = \sum_\lambda d_\lambda(X, t) \mathbf{P}_\lambda^t(z)$

where $a_\lambda(t)$, $b_\lambda(t)$, $c_{\nu,\lambda}(t)$, $d_\lambda(X, t)$ are rational functions in t . Explicit formulae known.

Examples:

- $\mathbf{P}_{[1,\dots,1]}^t(z) = \mathbf{e}_{[1,\dots,1]}(z)$
- $\mathbf{P}_{[s,\dots,s]}^t(z_1, \dots, z_r) = \prod_{i=1}^r z_i^s$, $\ell([s, \dots, s]) = r$

Determining conformal weights from the zero mode

We want to evaluate the zero mode of the singular vector

$$Q_+^{[p-1]}|(1-p)\alpha_+\rangle = \int_{\Gamma} S_+(z_1) \cdots S_+(z_{p-1})|(1-p)\alpha_+\rangle dz_1 \cdots dz_{p-1}.$$

The corresponding field is

$$\begin{aligned} Q_+^{[p-1]}V_{(1-p)\alpha_+}(w) \\ = \int_{\Gamma} S_+(z_1 + w) \cdots S_+(z_{p-1} + w)V_{(1-p)\alpha_+}(w) dz_1 \cdots dz_{p-1}. \end{aligned}$$

The easiest way to evaluate such a zero mode is to evaluate matrix elements

$$\langle \tau | Q_+^{[p-1]}V_{(1-p)\alpha_+}(w) | \tau \rangle,$$

where $\tau \in \mathbb{C}$ and $\langle \tau | a_{-n} = 0, n \geq 1$.

Determining conformal weights from the zero mode

$$\langle \tau | Q_+^{[p-1]} V_{(1-p)\alpha_+}(w) | \tau \rangle = \left\langle \mathbf{P}_{[(q-1)^{p-1}]}^{p/q}(z), \prod_{i=1}^{p-1} \left(1 + \frac{z_i}{w}\right)^{\alpha_+ \tau} \right\rangle_{p-1}^{p/q}$$

There are known formulae for expanding

$$\prod_{i=1}^{p-1} \left(1 + \frac{z_i}{w}\right)^{\alpha_+ \tau} = \sum_{\nu} g_{\nu}(\alpha_+ \tau) \mathbf{P}_{\nu}^{p/q}(z) w^{-|\nu|}.$$

Determining conformal weights from the zero mode

$$\langle \tau | Q_+^{[p-1]} V_{(1-p)\alpha_+}(w) | \tau \rangle \propto g_{[(q-1)^{p-1}]}(\alpha_+ \tau) \propto \prod_{r,s} (h_\tau - h_{r,s}),$$

where $1 \leq r \leq p-1$, $1 \leq s \leq q-1$, $(r, s) \simeq (p-r, q-s)$ and

$$h_\tau = \frac{\tau}{2}(\tau - \alpha_0), \quad h_{r,s} = \frac{(ps - qr)^2 - (p - q)^2}{4pq}.$$

Theorem

Let $f(h_\tau) = \prod_{r,s} (h_\tau - h_{r,s})$.

Zhu's algebra: $\text{Zhu}(\mathbf{M}(p, q)) = \frac{\mathbb{C}[x]}{\langle f(x) \rangle}$

The minimal model highest (conformal) weights are the zeros of $f(x)$.

Other algebras

$\mathcal{W}(p, q)$ triplet, $c = 1 - 6\frac{(p-q)^2}{pq}$, $p, q \geq 2$, $\gcd(p, q) = 1$.

$$\left\langle \mathbf{P}_{[(q-1)^{p-1}]}^{p/q}(z), \prod_{i=1}^{p-1} \left(1 + \frac{z_i}{w}\right)^{\alpha+\tau} \right\rangle_{p-1}^{p/q} = \prod_{r,s} (h_\tau - h_{r,s})$$

τ free boson charge; h_τ conformal weight; $r, s \in$ extended Kac table

Other algebras

$$\widehat{\mathfrak{sl}(2)}: k = -2 + \frac{p}{q}, p \geq 2, q \geq 1, \gcd(p, q) = 1.$$

$$\sum_{n=0}^{p-1} (-1)^n \binom{p-2-n+\sigma}{p-1-n} \sum_{m=0}^n \binom{p-1-m}{n-m} \\ \times \left\langle \mathbf{P}_{[q^m, (q-1)^{q-1-m}]}^{p/q}, \prod_{i=1}^{p-1} (1+z_i)^{-\sqrt{\frac{2q}{p}}\tau-1} \right\rangle^{p/q} \\ \propto g_p(\lambda_{\tau, \sigma}, h_{\tau}) \prod_{r,s} (h_{\tau} - h_{r,s}),$$

τ free boson charge; σ charge of $\beta\gamma$ ghosts; $\lambda_{\tau, \sigma}$ $\mathfrak{sl}(2)$ weight; h_{τ} conformal weight; $r, s \in \text{Kac table}$

Other algebras

$N = 1$ Virasoro: $c = \frac{3}{2} - 3\frac{(p-q)^2}{pq}$, $p, q \geq 2$, $p - q \in 2\mathbb{Z}$, $\gcd(p, \frac{p-q}{2}) = 1$.

$$\text{NS : } \left\langle \mathbf{P}_\delta^{-3}, \prod_{i=1}^{p-1} (1 + z_i)^{\sqrt{\frac{q}{p}}\tau} \right\rangle^{2p/(q-p)} \propto \prod_{r,s} (h_\tau - h_{r,s}^{(\text{NS})})$$

$$\text{R : } \left\langle \sum_\lambda c_\lambda \mathbf{P}_\lambda^{-3}, \prod_{i=1}^{p-1} (1 + z_i)^{\sqrt{\frac{q}{p}}\tau - \frac{1}{2}} \right\rangle^{2p/(q-p)} \propto \prod_{r,s} (h_\tau - h_{r,s}^{(\text{R})})$$

τ free boson charge; h_τ conformal weight; $r, s \in \text{Kac table}$

The End

Thank you for coming!